Speech in NLP: Text-to-Speech (TTS)

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August 9, 2026

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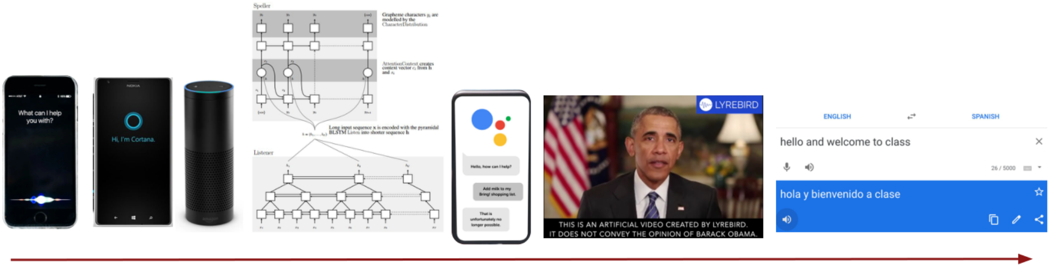
The power of human voice

"Words mean more than what is set down on paper.
It takes the human voice to infuse them with shades of deeper meaning."
— *Maya Angelou, I Know Why the Caged Bird Sings*

A decade of voice technology in products



2011: Apple Siri

2014: Microsoft Cortana, Amazon Alexa, Alexa Prize

2015: End-to-end neural becomes SORA

2016: Google assistant

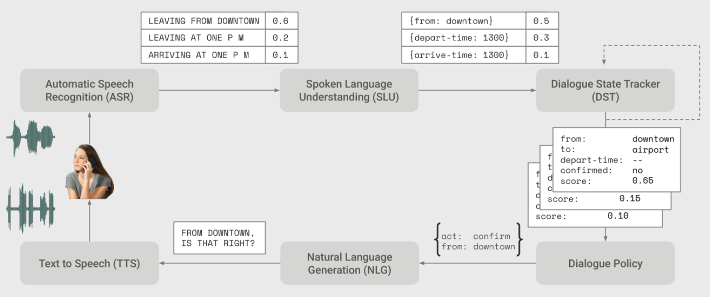
2017: Neural TTS voice cloning

2020: Realtime speech-speech translation

Source: Stanford CS224S, Lecture 1 (2024 Spring), slide 4. The 2015 label is a typo for SOTA, state of the art.

Where TTS sits: the spoken dialogue pipeline

- A task-oriented dialogue system recognizes speech, tracks what the user wants, decides what to say, and speaks the answer. TTS is the last stage, and the only one the user actually hears.



Source: Architecture of a dialogue-state system for task-oriented dialogue, after Williams et al. (2016); Stanford CS224S, Lecture 1 (2024 Spring), slide 19.

What is Text-to-Speech (TTS)?

- **What:** Produce speech from a text input (**Input:** text, optionally with speaker, language, style, or acoustic prompt; **Output:** waveform or codec tokens perceived as speech).
- **Applications:**
  - Accessibility (screen readers, assistive tech)
  - Personal Assistants (Apple Siri, Microsoft Cortana, Google Assistant)
  - Navigation, games, announcements/voice-overs, audiobooks, dubbing, AR/VR
  - Synthetic data for ASR

What is Text-to-Speech (TTS)?

- **Core difficulty:** Many-to-many mapping: the same text admits many natural realizations (prosody, rate, emphasis).
- **New variation:** Voice cloning: TTS systems that mimic particular speakers with minimal training data.

Prosody: Why the Same Text Sounds Different

- **Prosody** covers everything the words alone do not fix: intonation (the pitch contour, F_0), duration and rhythm, and energy (stress).
- Stress changes meaning: "**I** never said she stole my money" accuses differently than "I never said she stole **my** money". Seven words, seven readings, one spelling.
- Intonation signals sentence type: statements tend to fall in pitch, yes/no questions tend to rise.
- TTS systems model prosody implicitly (attention, latent variables) or explicitly (FastSpeech 2 predicts duration, pitch, and energy per phoneme).

TTS Overview

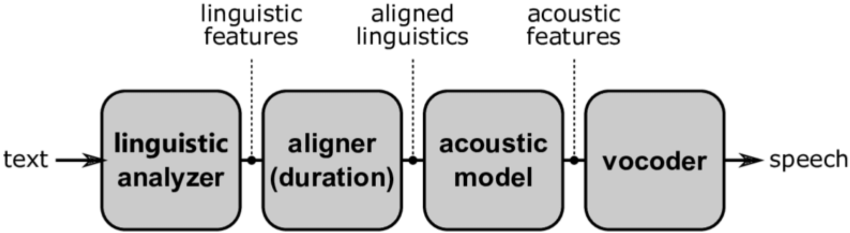
- Collect lots of speech (5–50 hours) from one speaker, transcribed carefully down to syllables and phones.
- Rapid recent progress in neural approaches.
- Modern systems are DNN-based, understandable, but not yet fully emotive.
- No longer require extensive data from a single speaker to make a TTS voice for them.

TTS: Definitions

- **Codec:** A coder/decoder system that encodes analog speech signals into a digitized, discrete compressed representation for compression.
- The discrete symbols a codec produces as its compressed form can serve as **discrete codes** for language modeling.
- A codec typically consists of:
  - An **encoder** (*e.g.*, convnets) that downsamples speech into an embedding,
  - A **quantizer** that converts the embedding into a series of discrete tokens,
  - A **decoder** (*e.g.*, convnets) that upsamples the tokens/embedding back into a (lossy) reconstructed waveform.

TTS: Modular Pipeline

1. **Text analysis:** normalization, tokenization, G2P
2. **Acoustic model:** mel, latents, or codec codes over time
3. **Vocoder / decoder:** waveform or neural codec inversion



Source: H.-T. Luong, "Deep learning based voice cloning framework for a unified system of text-to-speech and voice conversion," PhD thesis, SOKENDAI, 2020, Fig. 2.1.

Three eras of text-to-speech

- **Pre-2016:** concatenative unit selection and statistical parametric (HMM/GMM) synthesis.
- **2016–2021:** neural end-to-end models (Tacotron 2, WaveNet vocoders, FastSpeech, Glow-TTS, VITS).
- **2022–present:** diffusion, flow matching, and **foundation audio** models (VALL-E, Voicebox) with zero-shot cloning and in-context control.
- Each era targets bottlenecks of the previous one: flexibility, fidelity, latency, then controllability at scale.

Three eras of text-to-speech

Era	Technology	Strength	Bottleneck
Concatenative	Unit / diphone selection	Local timbre	Huge DB; brittle joins
Statistical (SPSS)	HMM / GMM vocoder params	Small footprint; adaptation	Muffled, averaged sound
Neural (E2E)	Tacotron 2 + neural vocoder	High fidelity; simpler stack	Slow AR inference
Foundation	VALL-E, Voicebox	Zero-shot; in-context learning	Data scale; misuse risk

Text Analysis: Normalization

- Raw text is full of **non-standard words**: numbers, dates, times, currency, abbreviations, URLs.
- The same string reads differently by context:
  - "Dr. Smith lives on King Dr." (doctor vs. drive)
  - "1995" as a year (nineteen ninety-five) or a quantity (one thousand nine hundred ninety-five)
  - "3/4" as a fraction (three quarters) or a date (March fourth)
- Classical solution: rules and lexicons per category; a large share of production TTS bugs start here.
- Modern systems use neural taggers or fold normalization into the model, but deployed stacks keep rule layers for safety.

Grapheme-to-Phoneme (G2P)

- English spelling underdetermines pronunciation: *though*, *tough*, *through*, and *thought* share letters, not sounds.
- **Homographs** need context to resolve: “I *read* it yesterday” vs. “I *read* every day”; *bass* the fish vs. *bass* the instrument.
- Standard pipeline: pronunciation lexicon lookup (CMUdict, about 134k entries written in ARPAbet, an ASCII phone alphabet for American English) plus a neural G2P model for out-of-vocabulary words and names.
- Phoneme input makes synthesis more stable; character-based models (Tacotron reads characters) learn G2P implicitly and can mispronounce rare words.

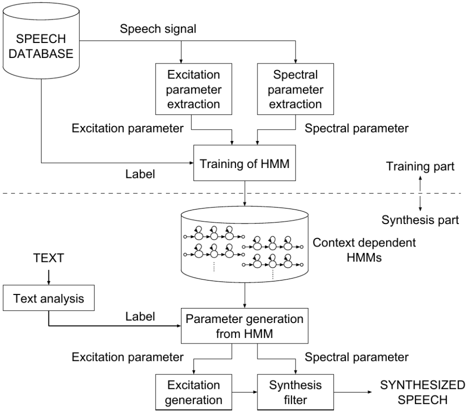
Pre-2016 Era: Corpus-Based and Concatenative Synthesis

- Early systems used **unit selection**: recording lots of speech from a single speaker and splitting it into small parts, usually syllables or **diphones**, a unit that runs from the middle of one phone to the middle of the next, so the hard part (the transition between two sounds) sits inside the recording rather than at a join.
- During synthesis, an algorithm picks which units to use by balancing:
  - **Target cost** (how well each piece matches the desired sound)
  - **Join cost** (how smooth the pieces sound when joined together)
- This gives good local naturalness, but is not very flexible and can sound glitchy if needed transitions are missing from the database.

Statistical Parametric Synthesis

- Statistical Parametric Speech Synthesis (SPSS) mainly used HMMs to model speech features instead of storing raw audio.
- SPSS models capture the spectral envelope, fundamental frequency (F_0), and duration of speech segments.
- Speaker adaptation is possible by updating a pre-trained model with a small amount of new speaker data, using maximum a posteriori (MAP) estimation or linear regression techniques.
- Trajectory HMMs were introduced to make transitions between features smoother and output speech more natural.

The HMM-based synthesis pipeline



Source: Tokuda, Zen & Black, "An HMM-Based Speech Synthesis System Applied to English," IEEE Workshop on Speech Synthesis, 2002, Fig. 1.

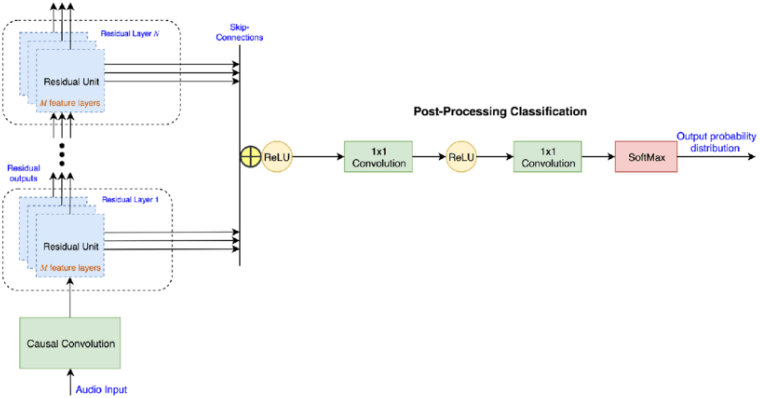
2016–2021 Era: The Neural Revolution (WaveNet)

- Deep generative model for raw audio
- Autoregressive: predicts next sample given previous samples
- High-quality, natural-sounding speech
- Equation:

$$p(x) = \prod_{t=1}^{T} p(x_t | x_1, \ldots, x_{t-1})$$

- Computationally expensive for real-time use

2016–2021 Era: The Neural Revolution (WaveNet)



Source: WaveNet architecture, redrawn by a third party (the paper's own Fig. 4 is a different, more compact drawing); architecture from van den Oord et al., "WaveNet: A Generative Model for Raw Audio," arXiv:1609.03499.

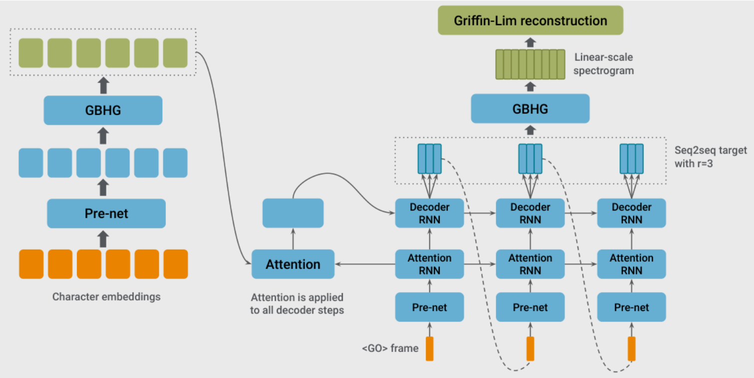
2016–2021 Era: Neural vocoders (WaveNet and beyond)

- Mel-spectrograms show how frequencies change over time, but must be turned back into sound waveforms.
- Griffin-Lim was the early answer: recover the missing phase from a magnitude spectrogram by iterating between the time and frequency domains. Cheap, but the quality was limited.
- Neural vocoders like WaveNet can reach human-level speech quality.
- WaveNet generates raw audio samples one at a time: 16,000 per second in the original paper, 24,000 in the Tacotron 2 vocoder.
- It uses dilated causal convolutions, so it can "see" a lot of context efficiently.
- WaveNet models the conditional probability of each sample:

$$p(\mathbf{x} \mid \mathbf{h}) = \prod_{t=1}^{T} p(x_t \mid x_1, \ldots, x_{t-1}, \mathbf{h})$$

  where $\mathbf{h}$ is the mel-spectrogram.
- This approach gives very natural speech, but it is slow to run for real-time use.

2016–2021 Era: End-to-end Neural TTS (Tacotron)

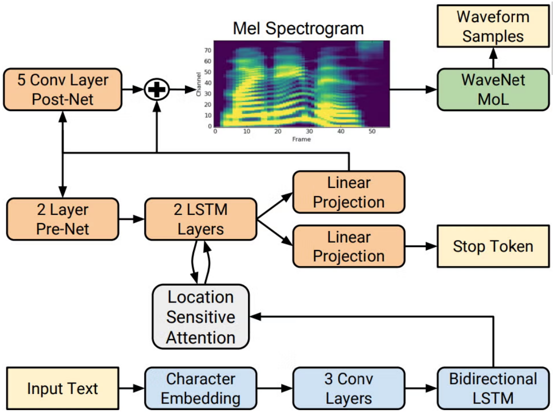


Source: Stanford CS224S, Lecture 1 (2024 Spring), slide 31, redrawing Wang et al., "Tacotron: Towards End-to-End Speech Synthesis," arXiv:1703.10135, Fig. 1. Characters in, raw spectrogram out, then Griffin-Lim to synthesize speech. The "GBHG" boxes are the slide's typo for CBHG (1-D convolution bank, highway network, bidirectional GRU).

Tacotron 2

- End-to-end TTS system
- Pipeline:
  - Text $\rightarrow$ Mel-spectrogram (sequence-to-sequence model)
  - Mel-spectrogram $\rightarrow$ WaveNet/GAN vocoder (audio synthesis)
- Improved prosody and pronunciation
- Flexible for different voices and languages

Tacotron 2



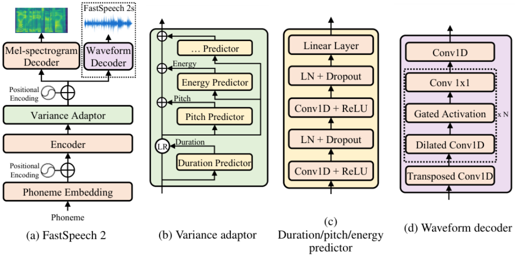
Source: Shen et al., "Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions" (Tacotron 2), Fig. 1; arXiv:1712.05884.

Autoregression: the deployment bottleneck

- **Latency:** Tacotron 2 with a WaveNet-class vocoder raised fidelity but blocked low-latency, on-device synthesis: one frame, then one sample, at a time.
- **Robustness:** the soft attention linking text to frames can also break down, so hard inputs come out with **skipped or repeated words**. Predicting an explicit duration per phoneme instead nearly removes those failures, and it is why later models search for a hard monotonic alignment rather than learning a soft one.
- Both problems point the same way: **non-autoregressive (NAR)** acoustic models and fast vocoders. FastSpeech reports mel generation 270$\times$ faster and end-to-end synthesis 38$\times$ faster than its autoregressive Transformer TTS teacher.

FastSpeech 2: parallel mel generation

- FastSpeech 2 uses a Feed-Forward Transformer with a Length Regulator to match phoneme and spectrogram lengths, replacing soft attention.
- The Variance Adaptor predicts duration, pitch, and energy for more control over prosody.
- Together they address the “one-to-many” mapping: the Duration Predictor decides exactly how many frames each phoneme lasts, which also makes the system more stable.
- Pitch Predictor models the fundamental frequency (F_0); Energy Predictor models the loudness (from the spectrogram).



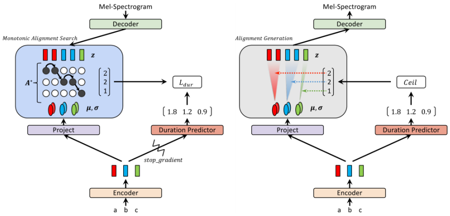
Source: Ren et al., “FastSpeech 2: Fast and High-Quality End-to-End Text to Speech,” Fig. 1; arXiv:2006.04558.

From noise to speech: flows, diffusion, and flow matching

- Once durations are predicted, the acoustic model only has to turn a noise sample into speech features. Three families do that, and they differ mainly in how many network passes one utterance costs.
- **Normalizing flow:** an invertible network maps speech features to a simple Gaussian. Training maximizes exact likelihood; generating runs the same network backwards on a noise sample, in a single pass. Used by Glow-TTS and inside VITS.
- **Diffusion:** a fixed forward process adds noise to the features until nothing is left but noise; the model learns to undo one step of it, so generating means denoising repeatedly, tens to hundreds of passes. Used by Grad-TTS.
- **Flow matching:** rather than a noising process, the model regresses the *velocity* along a straight path from noise to data. Generating integrates that velocity field with an ODE solver, which needs far fewer steps than diffusion for comparable quality. Used by Voicebox and F5-TTS.
- All three are conditioned on the text, so choosing between them is mostly a quality-versus-latency decision.

Glow-TTS and monotonic alignment search (MAS)

- Glow-TTS uses Normalizing Flows to turn a simple distribution into a mel-spectrogram distribution.
- Monotonic Alignment Search (MAS) finds the single best monotonic path between text and speech using dynamic programming.
- Unlike soft attention, MAS avoids alignment breaks, so the model works well on long utterances.

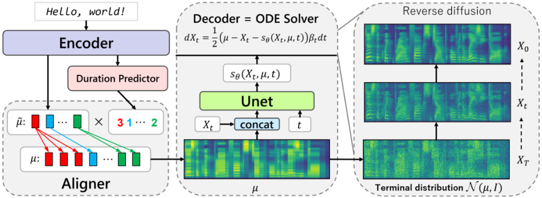


(a) An abstract diagram of the training procedure. (b) An abstract diagram of the inference procedure.

Source: Kim et al., "Glow-TTS: A Generative Flow for Text-to-Speech via Monotonic Alignment Search," Fig. 1; arXiv:2005.11129.

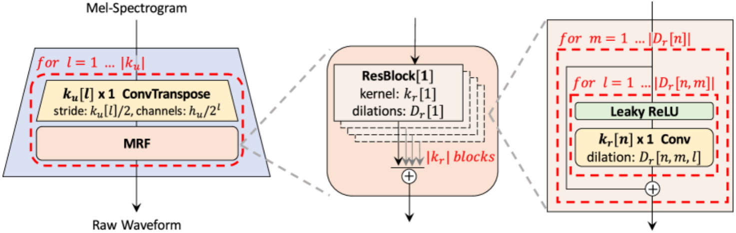
Grad-TTS: diffusion in the acoustic domain

- Grad-TTS uses score-based diffusion to turn random noise into structured mel-spectrograms.
- The reverse diffusion process is controlled by stochastic differential equations (SDEs).
- Users can control the trade-off between quality and speed by choosing the number of reverse steps.
- Grad-TTS can sound good with as few as 10 steps, and it is faster and more robust than Tacotron 2.



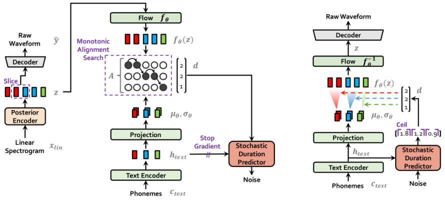
Source: Popov et al., "Grad-TTS: A Diffusion Probabilistic Model for Text-to-Speech," Fig. 2; arXiv:2105.06337.

- HiFi-GAN is a vocoder trained as a **generative adversarial network** (GAN): the generator turns mel-spectrograms into a waveform, discriminators try to tell its output from real recordings, and the generator improves by fooling them.
- The generator upsamples mel frames to waveform resolution with transposed convolutions; each stage passes through a multi-receptive field fusion (MRF) module that sums residual blocks with different kernel sizes and dilation rates, so short and long patterns are modelled together.
- Pairing it with a parallel front end like FastSpeech, which produces all mel frames at once, gives a fast and high-quality stack.



Source: Kong et al., "HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis," Fig. 1; arXiv:2010.05646.

- Trains as a conditional **variational autoencoder** (VAE): a posterior encoder maps real mel frames to a distribution over latent features and the waveform decoder reconstructs audio from a sample of it, so acoustic model and vocoder become one network.
- At training: monotonic alignment search (MAS) links text to speech, and a discriminator on the generated waveform supplies the adversarial loss.
- At inference: text $\rightarrow$ text encoder + duration predictor $\rightarrow$ latent features $\rightarrow$ waveform decoder $\rightarrow$ audio (no separate vocoder).

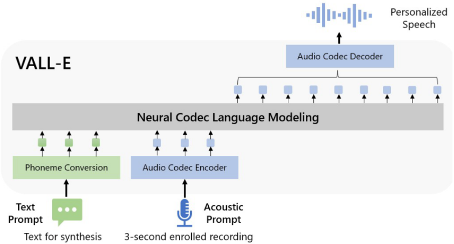


(a) Training procedure (b) Inference procedure

Source: Kim, Kong, and Son, "Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech" (VITS), Fig. 1; arXiv:2106.06103.

VALL-E: zero-shot TTS

- VALL-E changes the TTS pipeline to phoneme → discrete code → waveform, instead of phoneme → mel-spectrogram → waveform.
- It takes a text prompt and a short sample of the target voice, tokenizes both (grapheme-to-phoneme for the text, an audio codec for the speech), and uses a language model to generate discrete audio tokens for the text in that speaker's voice.
- VALL-E can do zero-shot TTS, speech editing, and content creation with other generative AI models like GPT-3.



Source: Wang et al., "Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers" (VALL-E), Fig. 1; arXiv:2301.02111.

VALL-E: comparison with other cascaded TTS systems

	Current Systems	VALL-E
Intermediate representation	mel spectrogram	audio codec code
Objective function	continuous signal regression	language model
Training data	$\leq$ 600 hours	60K hours
In-context learning	✗	✓

Source: Wang et al., VALL-E, Table 1: a comparison between VALL-E and current cascaded TTS systems; arXiv:2301.02111.

VALL-E: Limitations

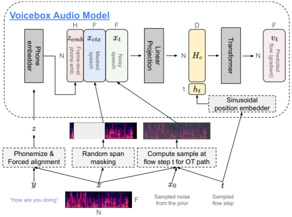
- **Data requirements:** VALL-E requires large amounts of high-quality paired text and audio data for effective training.
- **Speaker similarity:** The system relies on a reference sample; quality of voice cloning can degrade for voices and accents underrepresented in training.
- **Codec/token bottlenecks:** Using discrete audio tokens (from speech codebooks) may cause artifacts or limit audio fidelity compared to raw waveform modeling.
- **Generalization:** Performance may drop for out-of-distribution prompts or novel voice/text combinations not seen in training.
- **Ethical concerns:** Zero-shot voice synthesis can enable impersonation and raises privacy and misuse risks.

Voicebox: GPT-like generalist for speech

- Non-autoregressive flow-matching model trained to fill in missing speech given context audio and full transcript.
  - Style comes from audio context; content comes from text.
- One model handles zero-shot TTS, denoising, editing, style transfer, and diverse sampling via in-context learning.
- Gains over VALL-E on intelligibility and speaker similarity, up to **20**× faster generation.

Voicebox: Approach

- Training: learn $p(x_{\text{masked}} \mid y, x_{\text{context}})$, filling a random masked span using transcript y and unmasked audio.
- Two modules: an audio model (80-dim log-mel frames, Transformer) and a duration model.
- **Inference:** sample noise, then shape it into mel-spectrogram frames by solving a continuous flow ordinary differential equation (ODE) conditioned on the transcript and the audio context. A neural vocoder turns the predicted mel frames into a waveform.



Source: Le et al., Voicebox, Fig. 2: the Voicebox audio model; the lower half shows how training inputs are created; arXiv:2306.15687.

Voicebox: limitations

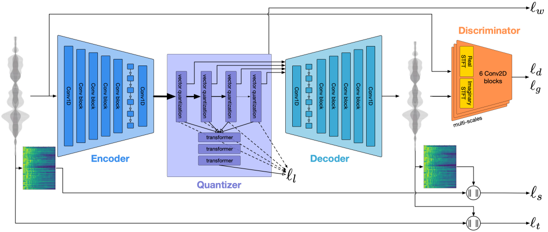
- **Read speech domain:** trained mainly on audiobooks; may not transfer well to casual conversation (laughter, back-channels).
- **Front-end dependency:** needs a phonemizer (the G2P front end) and a forced aligner to mark where each phoneme falls in the audio; word-based phonemizers miss context-sensitive pronunciation.
- **Limited control:** can transfer overall style from prompts, but cannot separately pick voice from one clip and emotion from another.
- **Bias risk:** performance can drop for underrepresented accents unless training data stays diverse.
- **Misuse:** high-quality cloning raises deepfake concerns; the authors report a highly effective synthetic-speech **classifier** as a mitigation, but neither it nor the model was released.

Comparing the Architectures

Model	Paradigm	Representation	Highlight
Tacotron 2	Autoregressive seq2seq	Mel	Strong naturalness
FastSpeech 2	NAR Transformer	Mel	Fast; explicit prosody
Glow-TTS	Flow + MAS	Mel	Hard alignments
Grad-TTS	Diffusion	Mel	Quality/speed trade-off knob
VITS	VAE + flow + GAN	Waveform (end-to-end)	No separate vocoder
VALL-E	Codec LM	Discrete codes	Zero-shot cloning
Voicebox	Flow matching	Mel (continuous)	Infilling / editing

Neural Audio Codecs: Audio as Tokens

- Neural audio codecs compress a waveform into a short sequence of **discrete tokens** and reconstruct audio from them: SoundStream (Zeghidour et al., 2021) and EnCodec (Défossez et al., 2022).
- An encoder downsamples the waveform, **residual vector quantization** maps each frame to entries of learned codebooks (each stage quantizes what the previous stage got wrong, so more codebooks means finer detail), and a decoder reconstructs the waveform from the codes.
- The result: audio becomes a token sequence, the same kind of object a language model already consumes and produces.



Source: EnCodec: encoder, residual vector quantizer, decoder, and discriminator. Défossez et al., "High Fidelity Neural Audio Compression," Fig. 1; arXiv:2210.13438.

Speech in the Language Model Pipeline

- Once audio is tokens, speech tasks become language modeling. VALL-E is the TTS instance: predict codec tokens conditioned on text and a short voice prompt.
- **AudioLM** (Borsos et al., 2022) showed pure audio continuation: a decoder-only model over audio tokens that continues speech or music with no text at all.
- **Speech-LLMs** put both directions in one model: ASR is audio tokens in, text out; TTS is text in, audio tokens out; the same transformer serves both.
- End-to-end voice conversation models take this to its conclusion: speech tokens in, speech tokens out, with no intermediate text transcript required.

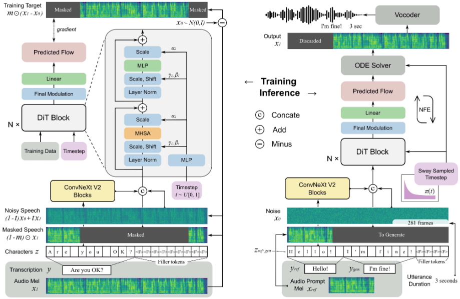
Why Token Pipelines Beat Cascades

- The classic voice assistant is a **cascade**: ASR $\rightarrow$ text LLM $\rightarrow$ TTS, three models and two conversions per turn.
- **Latency:** each stage waits for the previous one; a single token-level model can start responding while still listening.
- **Information loss:** a transcript discards tone, emphasis, emotion, and speaker identity; token pipelines keep them end to end.
- **Error compounding:** an ASR mistake becomes unrecoverable once only text is passed on.
- Cascades still win on modularity and controllability, which is why both designs coexist in production.

The 2026 TTS Landscape

- **Flow matching won the acoustic model:** F5-TTS and CosyVoice-class systems generate speech features non-autoregressively in a few ODE steps, with zero-shot cloning from a short reference clip.
- **Codec language models power conversation:** end-to-end voice modes speak through speech tokens directly, with no separate TTS stage.
- **Downloadable weights closed much of the gap:** XTTS, F5-TTS, and Kokoro all run locally, but check the licence before shipping: Kokoro is Apache-2.0, while the XTTS and F5-TTS checkpoints are non-commercial. Commercial systems (ElevenLabs, OpenAI, Google) still lead on expressiveness and robustness.
- Remaining hard problems: long-form consistency, fine-grained emotion control, low-resource languages, and singing.

- Text characters (padded with filler tokens) and a masked mel-spectrogram enter diffusion-transformer blocks; the model learns a flow from noise to the missing speech, conditioned on text and the audio prompt.
- Zero-shot cloning falls out of infilling, the same idea as Voicebox, with a simpler pipeline: no phonemizer, no duration model, no forced aligner.



F5-TTS training (left) and inference (right). Chen et al., "F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching," arXiv:2410.06885, Figure 1.

TTS in Real-Time Voice Agents

- A voice agent chains streaming ASR, an LLM, and TTS; the user hears silence until the first audio chunk arrives, so **time to first audio** matters more than total synthesis time.
- **Streaming synthesis:** start speaking while the LLM is still writing, chunking at sentence or phrase boundaries; codec-LM stacks can stream token by token.
- **Barge-in:** users interrupt mid-sentence; the stack must stop playback, flush queued audio, and hand the conversation back to the LLM.
- Expressive control is moving from Speech Synthesis Markup Language (SSML) tags toward natural-language style prompts ("speak warmly, a little slower").

Evaluating TTS quality

- **Human tests:** Does the voice sound natural?
- **Similarity:** Is the generated speech close to real speech?
- **Intelligibility:** Can speech-to-text systems understand it?
- **Cloning:** How well does it copy a specific voice?
- TTS is evaluated by playing a sentence to human listeners and having them give a mean opinion score (MOS) on a scale of 1 to 5.

Objective Evaluation

- Human MOS panels are slow and expensive, so large-scale development leans on objective proxies.
- **Intelligibility:** transcribe the synthesized speech with a strong ASR model and compute WER against the input text.
- **Speaker similarity:** cosine similarity between speaker embeddings of the reference prompt and the generated audio.
- **Predicted MOS:** neural predictors (UTMOS, NISQA) estimate human scores from audio alone.
- Modern systems score close to human recordings, so plain MOS saturates; side-by-side comparative MOS (CMOS) separates systems better.

Voice Cloning Safeguards

- **Consent and disclosure:** clone only with the speaker's permission and label synthetic audio as synthetic.
- **Watermarking:** embed an inaudible signature in generated audio so it can be identified later (AudioSeal).
- **Detection:** train classifiers to spot synthetic speech; the Voicebox authors report a highly effective classifier for their own outputs.
- These mitigations are partial: watermarks do not survive every re-encoding, detectors lag the newest generators, and neither the Voicebox model nor its detector was publicly released.

Ethics before you collect speech

أكاديمية كاوست
KAUST ACADEMY

Don't record someone without their consent

In California, all parties to any confidential conversation must give their consent to be recorded. For calls occurring over cellular or cordless phones, all parties must consent before a person can record, regardless of confidentiality.

Don't create a speech synthesizer / voice clone of someone without their consent

It might be fun, but it's a little creepy. People get upset.

Okay to use existing speech datasets (we'll provide some).

Do consider subgroup and language bias when building systems

Poor performance on subgroups e.g. non-native speakers

Many languages are under-served relative to English/Mandarin.

Source: Stanford CS224S, Lecture 1 (2024 Spring), slide 8.

Takeaways

- One text can be spoken many ways (speed, stress, intonation). Most systems still go text $\rightarrow$ acoustic features $\rightarrow$ waveform; newer models like VALL-E predict audio tokens directly.
- TTS improved in stages: stitch recorded clips $\rightarrow$ slow but high-quality neural models (Tacotron 2, WaveNet) $\rightarrow$ faster parallel models (FastSpeech 2, HiFi-GAN, VITS) $\rightarrow$ foundation models that clone voices from a short sample (VALL-E, Voicebox).
- Voice cloning from a few seconds of audio works well today, but can fail on rare accents; use it carefully with consent, labeling, and fake-speech detection.

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